Association for the Advancement of Artificial Intelligence

STProtein: predicting spatial protein expression with multi-omics data

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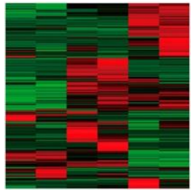
Outline

1. Background
2. Related Work
3. Methodology
4. Results
5. Summary

Background | Integration of AI & spatial omics brings new opportunity

Omics sequencing technology

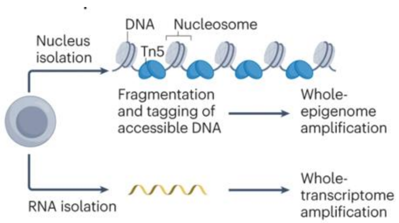


Transcriptome sequencing

2000~2010

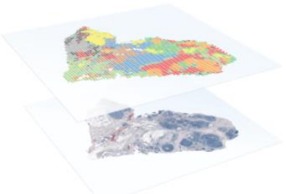
Single-cell transcriptome sequencing

2009



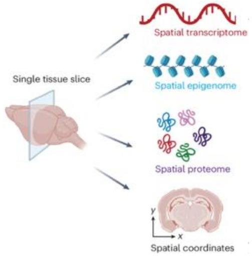
Single-cell multi-omics sequencing

2015



Spatial transcriptome sequencing

2019 (10X Visium)



Spatial multi-omics sequencing

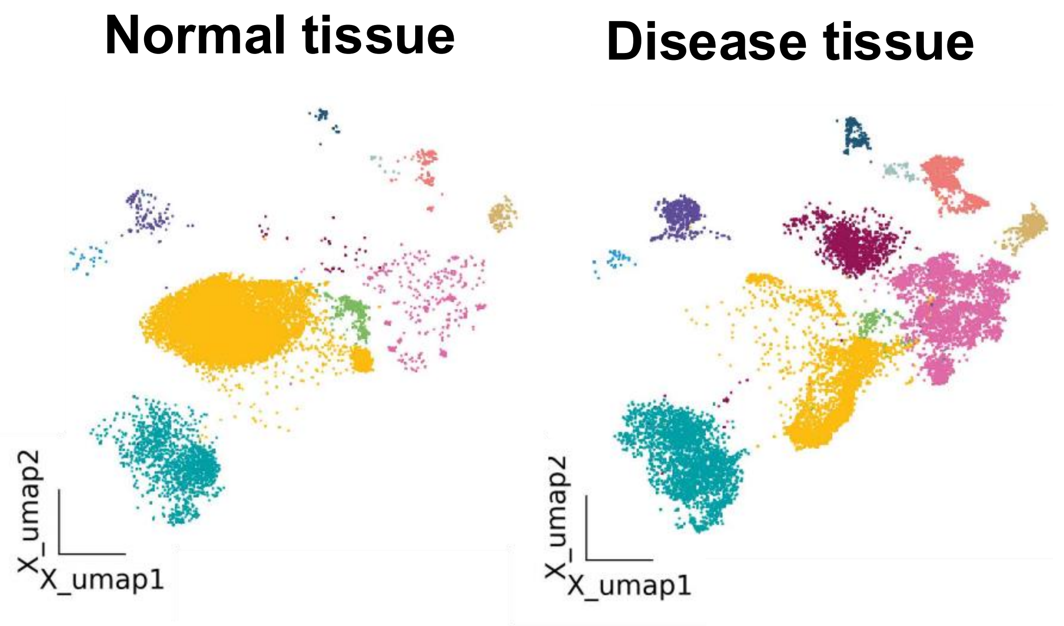
2023

The development of high-throughput sequencing technology has promoted the development of single-cell omics and spatial omics

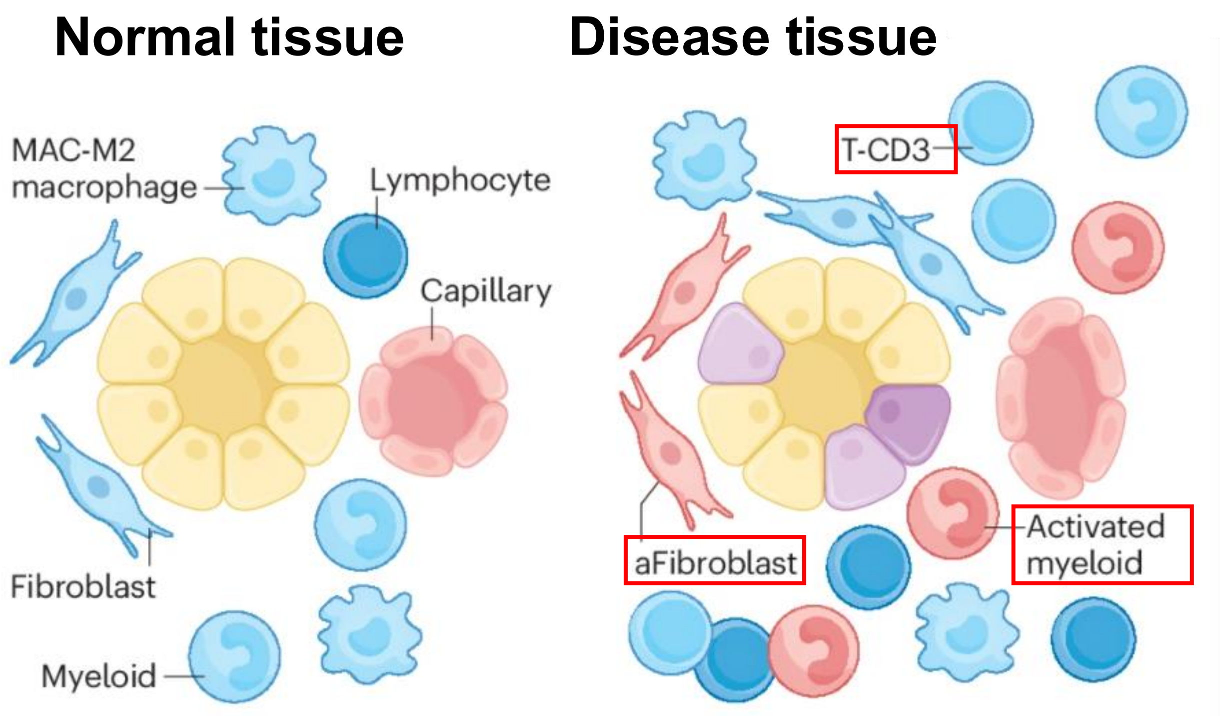
It provides the necessary data foundation and enabling opportunities for AI FOR LIFE SCIENCE!

Background | Boots studying of tumor immunity & cell development

Compared with single-cell omics, spatial omics can analyze the cellular composition and cell interactions of tissue microenvironment



Loss Spatial Information



Cellular composition and cell interactions affect cell phenotype (normal or disease)

Jain, S., Eadon, M.T. ***Nature Reviews Nephrology***, 2024

Related Work | Existing methods not considered spatial information

Cell type

Regulatory relationships

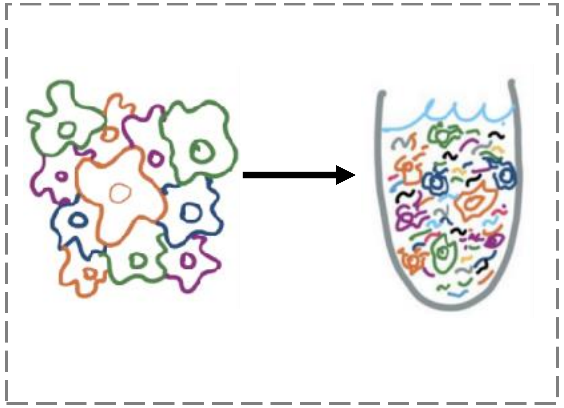
Spatial information

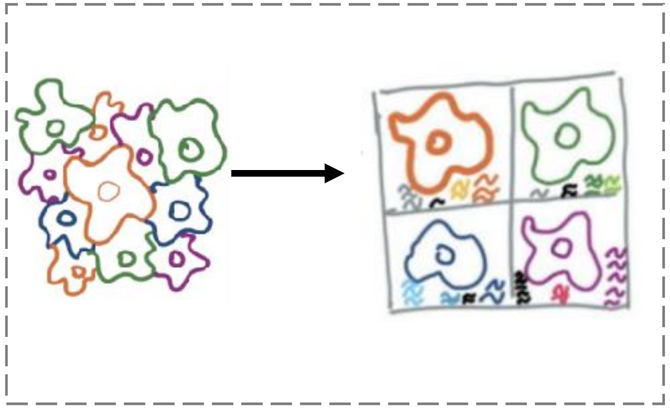
Transcriptome sequencing

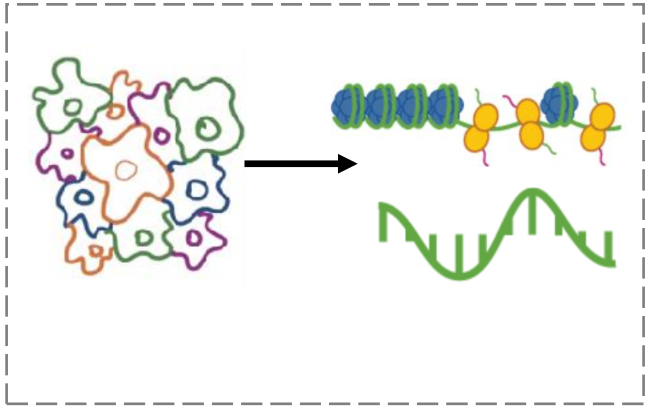
Single-cell transcriptome sequencing

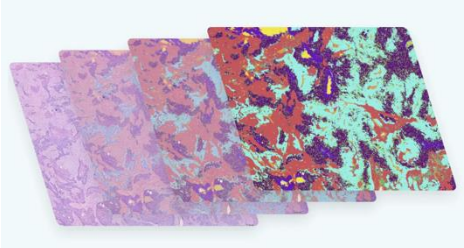
Single-cell multi-omics sequencing

Spatial multi-omics sequencing





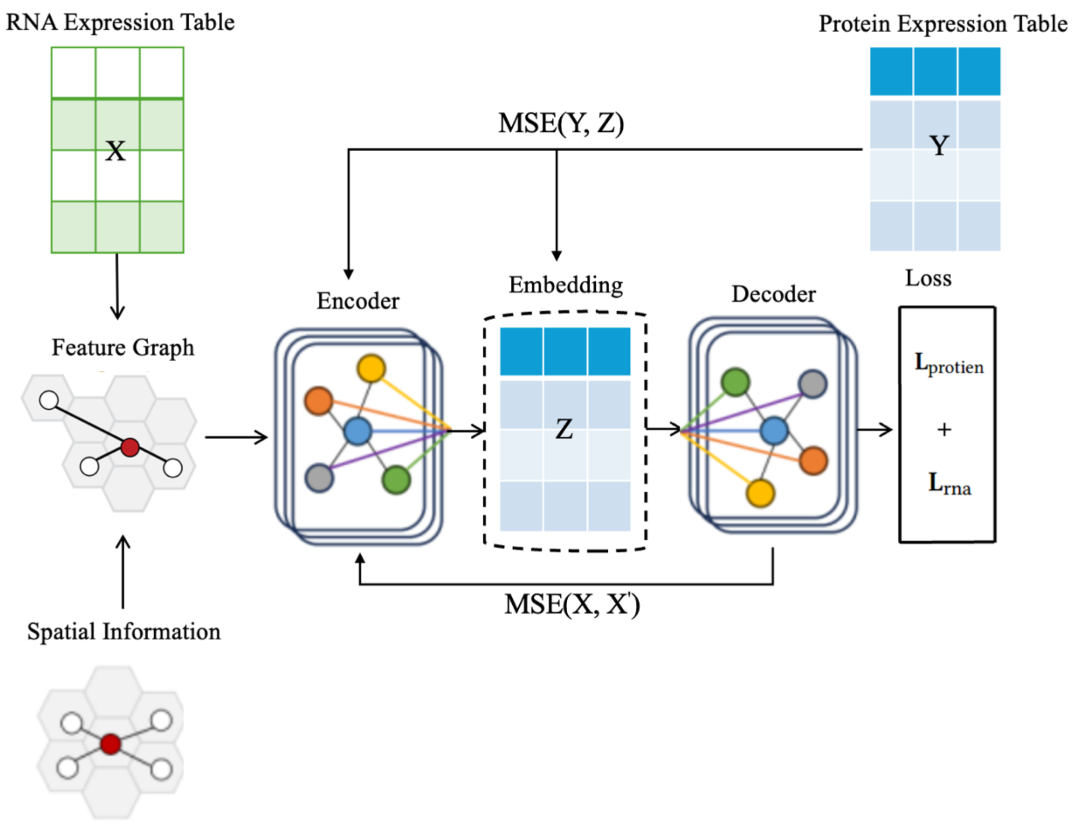




Lack of computing tools

Lack of computational models for prediction of missing data in spatial multi-omics!
There is a serious "imbalance" between spatial transcriptomic and proteomic data!
Compared with spatial transcriptome sequencing, spatial proteome sequencing is much more expensive!

Methodology | STProtein Training Framework



Methodology：

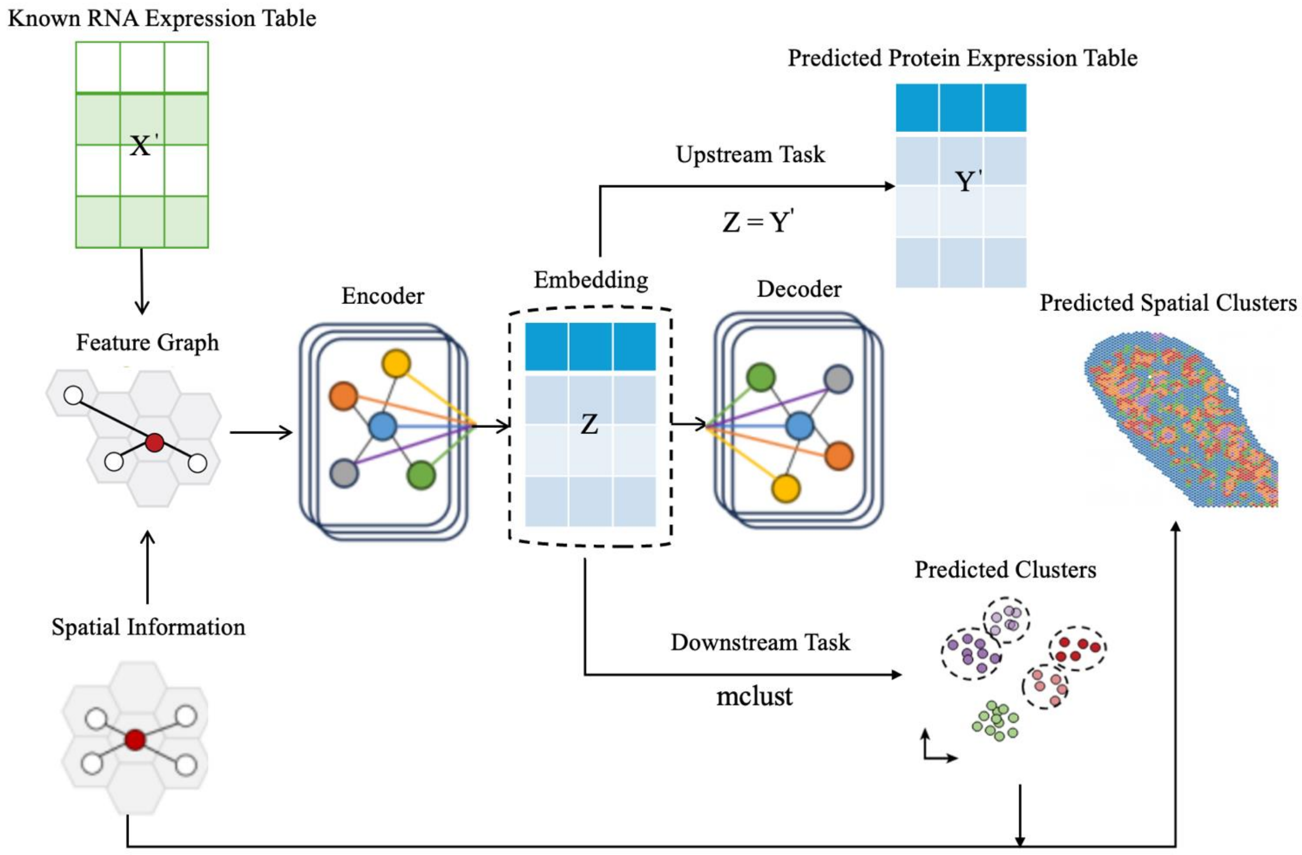
Representation of spatial transcriptomics

Graph Autoencoder and Attention

Latent embeddings represent the spatial proteomic profiles

Using graph autoencoders to map spatial transcriptomics to spatial proteomics

Methodology | STProtein: Upstream and downstream pipelines



Methodology：

Based on pre-trained STProtein model

Prior spatial transcriptomic information

Newly obtained embeddings facilitate upstream and downstream tasks

Inferring unknown spatial proteomics from prior spatial transcriptomic data

Results | Upstream Task Evaluation

Table 4. Dataset's detialed information for STProtein

Name	Platform	Size(spots x genes/proteins)
Mouse spleen replicate 1	SPOTS (RNA-protein)	2,568x32,285 2,568x21
Mouse spleen replicate 2	SPOTS (RNA-protein)	2,768x32,285 2,768x21
Mouse Thymus 1	Stereo-CITE-seq (RNA-protein)	4,697x23,622 4,697x51
Mouse Thymus 2	Stereo-CITE-seq (RNA-protein)	4,253x23,529 4,253x19
Mouse Thymus 3	Stereo-CITE-seq (RNA-protein)	4,646x23,960 4,646x19
Mouse Thymus 4	Stereo-CITE-seq (RNA-protein)	4,228x23,221 4,228x19
Human Lymph Node A1	10x Genomics Visium (RNA-protein)	3,484x18,085 3,484x31
Human Lymph Node D1	10x Genomics Visium (RNA-protein)	3,359x18,085 3,359x31

Table 3. Spatial transcriptomics and proteomics data format

Data type	Format	Characteristic
Spatial Transcriptomics	RNA Expression Table	Rows represent genes; columns represent cells; value indicates RNA expression level
	Spatial Information	Contains coordinates $(\bar{x}, \bar{y})$ for each cell in the tissue
Spatial Proteomics	Protein Expression Table	Rows represent genes; columns represent cells value indicating protein expression level
	Spatial Information	Contains coordinates $(\bar{x}, \bar{y})$ for each cell in the tissue

Table 6. Quantitative Upstream Evaluation on Three Datasets by Using RMSE Metric

Methods	Mouse Spleen Dataset	Mouse Thymus Dataset	Human Lymph Node Dataset
totalVI	1.05	1.42	1.22
scArches	1.03	1.38	1.20
Dengkw	0.99	1.05	1.17
cTp_net	1.27	1.47	1.27
STProtein	**0.95**	**0.98**	**1.00**

STProtein delivers the most accurate protein expression predictions across mouse spleen, mouse thymus, and human lymph node datasets

Results | Downstream Task Evaluation.

Table 7. Quantitative Downstream Evaluation on Mouse Spleen Dataset (SPOTS)

Methods	NMI(%)	AMI(%)	FMI(%)	ARI(%)	V-Measure(%)	F1-Score(%)	Jaccard(%)
totalVI	25.58	25.39	41.49	4.84	25.58	39.42	24.55
scArches	26.40	26.21	43.28	5.28	26.40	40.61	25.47
Dengkw	28.69	28.52	39.66	8.19	28.69	39.46	24.58
cTp_net	17.22	17.04	42.33	18.39	17.22	42.27	26.80
STProtein	**30.91**	**30.75**	**60.35**	**40.43**	**30.91**	**59.74**	**42.59**

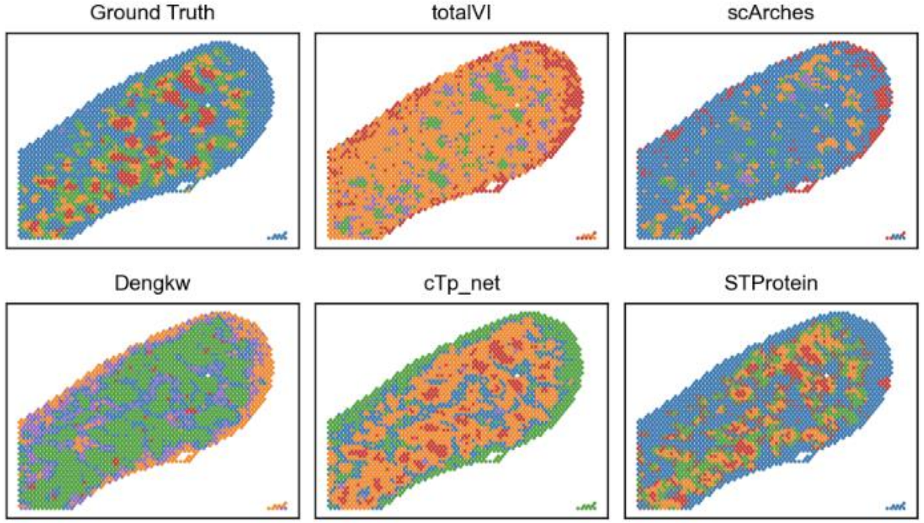


Fig. 5. Visualization Results about Comparison of Benchmarking Prediction Methods and STProtein with Original Ground Truth on Mouse Spleen Dataset (SPOTS)

Table 9. Quantitative Downstream Evaluation on Human Lymph Node Dataset (10x Genomics Visium)

Methods	NMI(%)	AMI(%)	FMI(%)	ARI(%)	V-Measure(%)	F1-Score(%)	Jaccard(%)
totalVI	29.90	29.70	51.56	21.63	29.90	50.63	34.10
scArches	41.71	41.55	52.80	29.12	41.71	52.77	35.84
Dengkw	41.18	41.00	55.75	35.67	41.18	56.73	40.57
cTp_net	35.66	35.48	48.13	24.42	35.66	48.11	31.67
STProtein	**42.31**	**42.12**	**56.82**	**35.97**	**42.31**	**57.22**	**41.26**

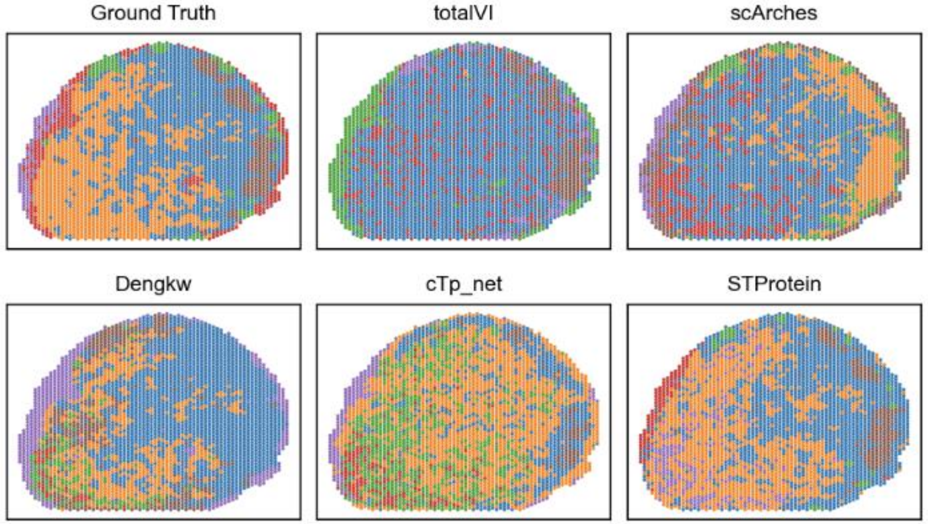


Fig. 7. Visualization Results about Comparison of Benchmarking Prediction Methods and STProtein with Original Ground Truth on Human Lymph Node (10x Genomics Visium)

STProtein outperforms existing prediction methods in spatial proteomic, clustering issue area on mouse spleen/thymus and human lymph node data

Results | Benchmarking Different Graph Convolutional Layer

Table 10. Prediction Results of Different Graph Convolutional Layer on Mouse Spleen Dataset (SPOTS)

Methods	RMSE
GCN	0.98
SAGE	0.96
Transformer	0.96
GAT	0.97
GATv2	**0.95**

Table 11. Clustering Results of Different Graph Convolutional Layer on Mouse Spleen Dataset (SPOTS)

Methods	NMI(%)	AMI(%)	FMI(%)	ARI(%)	V-Measure(%)	F1-Score(%)	Jaccard(%)
GCN	21.46	21.24	38.69	12.12	29.04	37.65	23.19
SAGE	29.24	29.65	44.60	22.52	29.24	42.08	26.64
Transformer	24.13	23.91	41.20	11.37	24.13	40.91	25.71
GAT	21.95	21.73	41.70	15.30	21.95	40.85	25.66
GATv2	**30.91**	**30.75**	**60.35**	**40.43**	**30.91**	**59.74**	**42.59**

GATv2 optimal results for both upstream and downstream tasks

Results | Ablation Studies

Table 12. Ablation study for feature graph construction and loss function design on prediction task

Methods	RMSE
STProtein	**0.95**
SN Feature Graph	0.96
$\mathbf{L}_{protien}$	0.97
$\mathbf{L}_{rna}$	1.02

Table 13. Ablation study for feature graph construction and loss function design on clustering task

Methods	NMI(%)	AMI(%)	FMI(%)	ARI(%)	V-Measure(%)	F1-Score(%)	Jaccard(%)
STProtein	**30.91**	**30.75**	**60.35**	**40.43**	**30.91**	**59.74**	**42.59**
SN Feature Graph	27.05	26.89	44.49	21.31	27.05	42.72	27.16
$\mathbf{L}_{protien}$	29.10	28.94	43.43	19.12	29.10	42.06	26.63
$\mathbf{L}_{rna}$	20.56	20.38	37.85	9.23	20.56	37.20	22.85

The KNN feature graph and the multi-task learning-based loss design prove highly effective for both upstream and downstream tasks

Results | Parameter Sensitivity Experiments

Table 14. Parameter sensitivity experiments for k value in KNN graph construction on prediction task

K value	RMSE
1	0.96
2	0.97
3	**0.95**
4	0.96
5	0.96

Table 15. Parameter sensitivity experiments for k value in KNN graph construction on clustering task

K value	NMI(%)	AMI(%)	FMI(%)	ARI(%)	V-Measure(%)	F1-Score(%)	Jaccard(%)
1	29.27	29.11	47.85	23.19	29.27	47.10	30.81
2	26.00	25.84	42.66	18.22	26.00	41.26	25.99
3	**30.91**	**30.75**	**60.35**	**40.43**	**30.91**	**59.74**	**42.59**
4	28.52	28.35	52.76	30.24	28.52	51.92	35.06
5	27.49	27.32	56.02	34.22	27.49	55.41	38.32

Table 16. Parameter sensitivity experiments for clustering algorithm choices on clustering task

Algorithms	NMI(%)	AMI(%)	FMI(%)	ARI(%)	V-Measure(%)	F1-Score(%)	Jaccard(%)
mclust	**30.91**	**30.75**	**60.35**	**40.43**	**30.91**	**59.74**	**42.59**
leiden	28.36	28.19	51.43	28.06	28.36	50.71	33.97
louvain	27.29	27.12	53.08	30.20	27.29	52.42	35.52

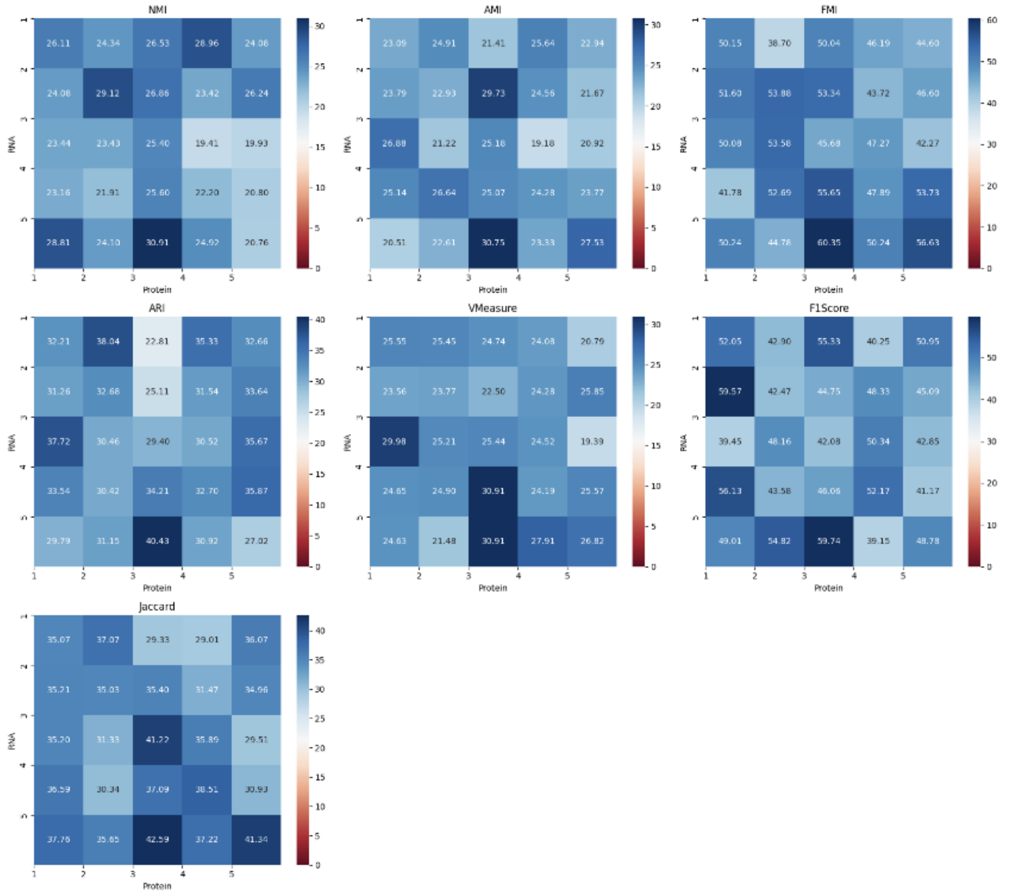


Fig. 4. Parameter Sensitivity Experiments for Reconstruction Loss Weights for RNA and Protein Items on Mouse Spleen Dataset (SPOTS). The Data Presented in Heatmap are Presented in Percentages.

The configuration yields superior results with a K=3 as KNN feature graph, mclust clustering, and a 5:3 loss weight ratio

Results | Scientific Discovery of 'Dark Matter'

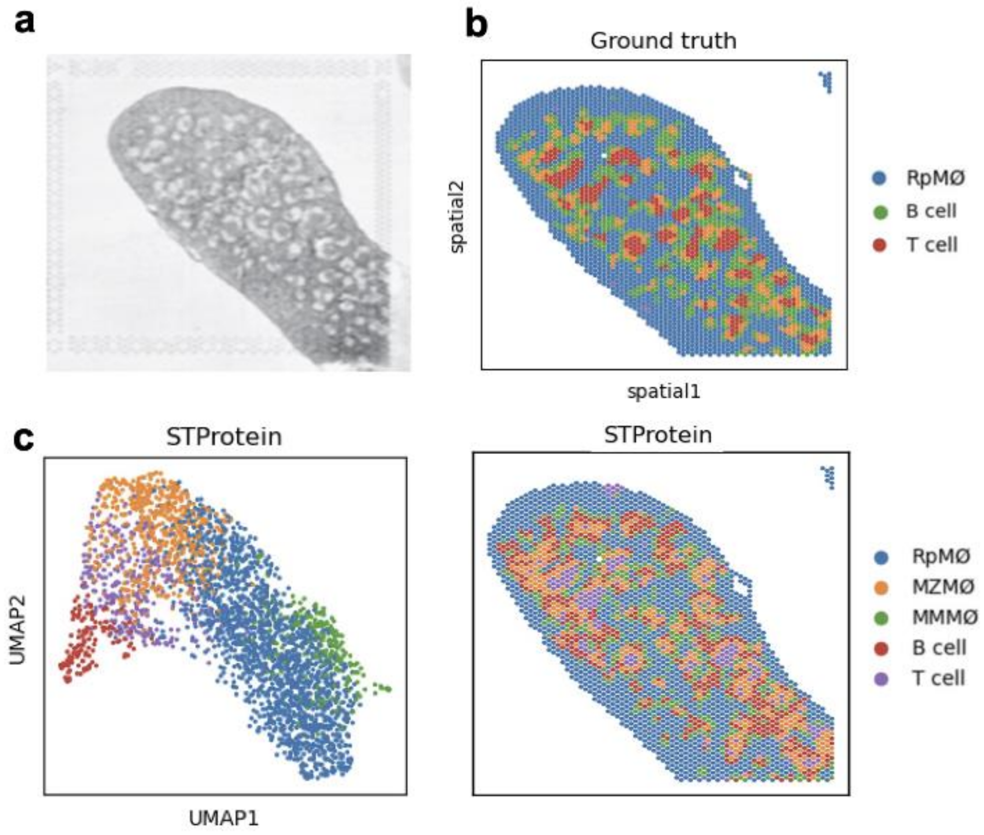


Fig. 8. a): H&E Histological Image for Mouse Spleen Structure; b): Ground Truth of Clustering Results for Mouse Spleen with its Orignial Annotation (RpMZΦ, B Cell and T Cell) Shown in the Right.; c): UMAP Picture and clustering Visualization Picture for STProtein with its Annotation (RpMZΦ, MZMΦ, MMMΦ, B Cell and T Cell) Shown in the Right.

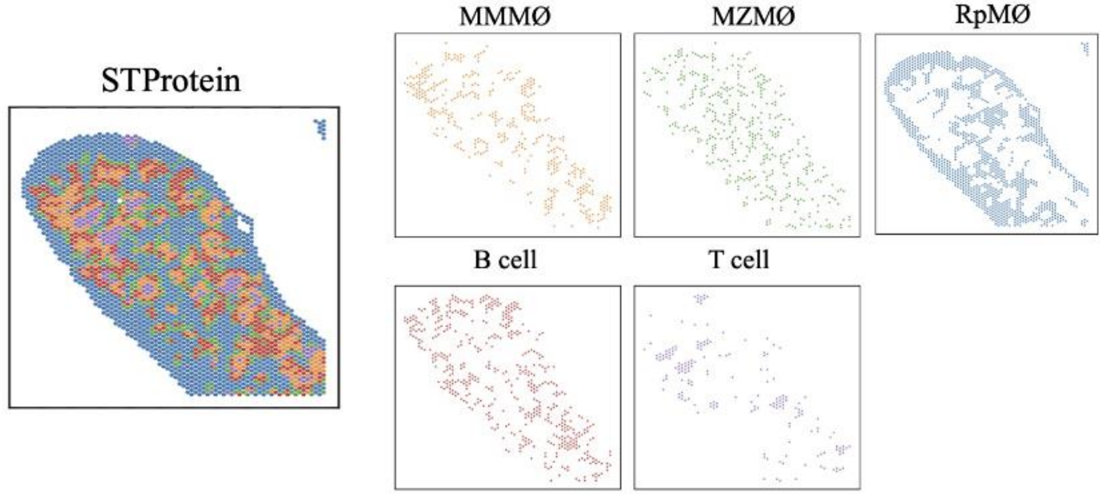


Fig. 11. The New Annotation of Marker Genes on Mouse Spleen Dataset (SPOTS) by Using STProtein

Acting as a 'Telescope' for researchers, STProtein uncovers 'Dark matter' in unexplored realms. This accelerates scientific discovery, which lies at the heart of AI4Science

STProtein identifies previously unannotated "Dark Matter" tissues, surpassing the limitations of manual labelling

Summary

- Design spatial multi-omics integration algorithms that integrate spatial information for spatial multi-omics data
- Develop spatial proteomics prediction algorithms using multi-task learning for spatial multi-omics data
- Provide new tools for spatial multi-omics scientific research by discovery “Dark Matter” powered by deep learning

GitHub: https://github.com/zhaorui-bi/STProtein

Contact: Zhaorui JIANG (zrjing25@stu.pku.edu.cn)

THANK YOU FOR YOUR LISTENING